BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 9221

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 9221

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The process of matching two figures is called .....

(A) congruent
(B) duplicate
(C) superposition
(D) concurrent

Q2. Medians of a triangle are divided by the point of concurrency in the ratio .....

(A) 2:1
(B) 2:3
(C) 1:3
(D) none of these

Q3. Which of the following are concurrent?

(A) angle bisectors of a triangle
(B) perpendicular bisectors of the sides of a triangle
(C) medians of a triangle
(D) all of these

Q4. A diagonal matrix in which all the diagonal elements are equal is said to be ..... matrix.

(A) null
(B) scalar
(C) square
(D) diagonal

Q5. LCM =.....

(A) HCF/A×B
(B) B/HCF
(C) A/HCF
(D) A×B/HCF

Q6. $\sqrt{(2x+1)}-5=4$ is a ..... equation

(A) linear
(B) radical
(C) cubic
(D) quadratic

Q7. The line y=b is parallel to .....

(A) x-axis
(B) y-axis
(C) both A&B
(D) none of these

Q8. Coordinates of mid-point of the line joining A(5,−4) and B(−6,5) are .....

(A) (−1,1)
(B) (11,9)
(C) (½,½)
(D) (−½,½)

Q9. For which of the following expressions a+b is not a factor?

(A) a^2-b^2
(B) a^2+b^2
(C) a^3+b^3
(D) a^4-b^4

Q10. In simplified form $1/(a+b) + b/(a^2-b^2)$ = ...

(A) $(b+1)/(a^2-b^2)$
(B) $a/(a^2-b^2)$
(C) $b/(a^2-b^2)$
(D) $(b+a)/(a^2-b^2)$

Q11. If length and width of a rectangle are x+18 and x−5 respectively then perimeter P is equal to .....

(A) (x+18)(x−5)
(B) 2x+13
(C) 4x+26
(D) 0−23

Q12. A square matrix A is called non-singular if .....

(A) $|A|\neq 0$
(B) $|A|=0$
(C) $A^t=0$
(D) $|A^t|=0$

Q13. If n=8 and $16\times 2^m = 4^{n-8}$, then m = ...

(A) 8
(B) −2
(C) 0
(D) −4

Q14. Characteristic of log 4.35 is .....

(A) 0
(B) 1
(C) 2
(D) 3

Q15. ..... is the only number which can not be represented by Roman numerals.

(A) 50
(B) 10
(C) 0
(D) 100

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 36 | **Paper Code:** 9221

Q1. Attempt any nine of the following short questions. Each question carries 4 marks.

(i) If A=[0 −1; 2 1] and B=[2 3; 1 0] then show that $(AB)^{-1} = B^{-1}A^{-1}$

(ii) Simplify (238.2)(9.506) with the help of logarithm. (Using log table)

(iii) Simplify $\sqrt{(625x^3y^4/25xy^2)}$

(iv) Find the value of x+y+z when $x^2+y^2+z^2=10$ and xy+yz+zx=11

(v) Factorize $(t+1)^2+3(t+1)+2$

(vi) Prove that (x+2) is a factor of x^3-7x-6 and hence find the other factors.

(vii) If product of two polynomials is $x^4+5x^3-6x^2-2x-28$ and their HCF is x−2. Find their LCM

(viii) Simplify $(9-x^2)/(x^4+6x^3) \div (x^3-2x-3x)/(x^2+7x+6)$

(ix) If breadth of a room is one fourth of its length and the perimeter of the room is 20 m. Find the length and breadth of the room.

(x) Solve the following system of equations graphically: y=2x+2, y=x−1

(xi) The mid point and one end of a line segment are (3,7) and (4,2) respectively. Find the other end point.

(xii) Solve the system of linear equations using Cramer's rule: 5x+7y=3, 3x+y=5

SECTION-C

Maximum Marks: 24

> Note: Attempt any three of the following questions. Each question carries 8 marks.

Q2. Prove that the points A(−2,−2), B(4,−2) and C(4,6) are vertices of a right triangle.

Q3. Show that from a point outside a line, the perpendicular is the shortest distance from the point to the line.

Q4. Prove that parallelograms on equal bases and having same altitude are equal in area.

Q5. Construct ΔUVW, draw their altitudes and verify their concurrency: mUV=4.5cm, mVW=3.9cm and m∠W=45°